ZCAS UNIVERSITY

BACHELOR OF SCIENCE IN NETWORK ENGINEERING

CCS3622 TELECOMMUNICATION AND WAN

FINAL EXAMINATION

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

DATE : 25th NOVEMBER, 2024

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. Question 2 is practical. When attempted, submit the soft copy to the invigilator.
4. This examination paper carries a total of **100 marks.**
5. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A – COMPULSORY

Study the description below:

QUESTION 1

The foundation of any organization is communication. Companies must take into account communication between a number of individuals, such as staff members and supervisors, consumers and clients, and suppliers or other external entities. The instruments and techniques employed in communication have a big influence on output, quality, and outcomes. Without the proper tools, communication is probably going to be far less effective and efficient.

For any organization, communication and connectivity are essential. For businesses to stay relevant and compete with other organizations, they must be able to deliver information quickly, thoroughly, and flexibly. Using telecommunication technology allows organizations to stay competitive by enabling them to speak directly with customers, obtain information at any time and from any location, stay up to date with their market, and build relationships and resources. Telecommunications and information technology have become essential components of society, including business operations. Nowadays, communications is necessary to carry out even the most fundamental company operations and procedures smoothly and in a way that keeps them competitive. It has fundamentally changed the way the business world functions, and all contemporary companies must make sure they have

These networks are frequently created by service providers, who subsequently rent their wide area network (WAN) to the general public, corporations, governments, and educational institutions. As long as they have access to the established WAN, these clients can use the network to connect with other users or relay and store data, regardless of where they are. Access can be provided by a variety of channels, including internet access, cellular networks, wireless networks, and virtual private networks (VPNs) or lines. WANs enable worldwide firms to quickly and efficiently perform their vital everyday tasks. Workers can connect to the company's larger data resource center, share data, and communicate with colleagues from any location by using the wide area

network (WAN). Organizations can maintain their internal wide area network with the assistance of certified network professionals.

Having read the scenario above, answer the following questions:

a) Explain four (4) benefits of implementing telecommunication and WAN networks in business organizations. (8 marks)

b) Discuss any three (3) challenges of wide area networks (WANs). (6 marks)

c) Describe the following equipment and terms used on Wide Area Networks (WANs)

i) Data terminal equipment (DTE) (3 marks)

ii) Data Circuit-terminating Equipment (DCE) (3 marks)

iii) Demarcation point (demarc) (3 marks)

iv) Local loop/ Last mile (3 marks)

v) Customer Premises Equipment (CPE) (2 marks)

d) One of the greatest challenges for network technicians is to maintain network or system running 24/7. Network downtime should be minimized as much as possible and increase uptime. Outline three (3) mechanisms a network engineer can implement to ensure networks are operational 24/7. (6 marks)

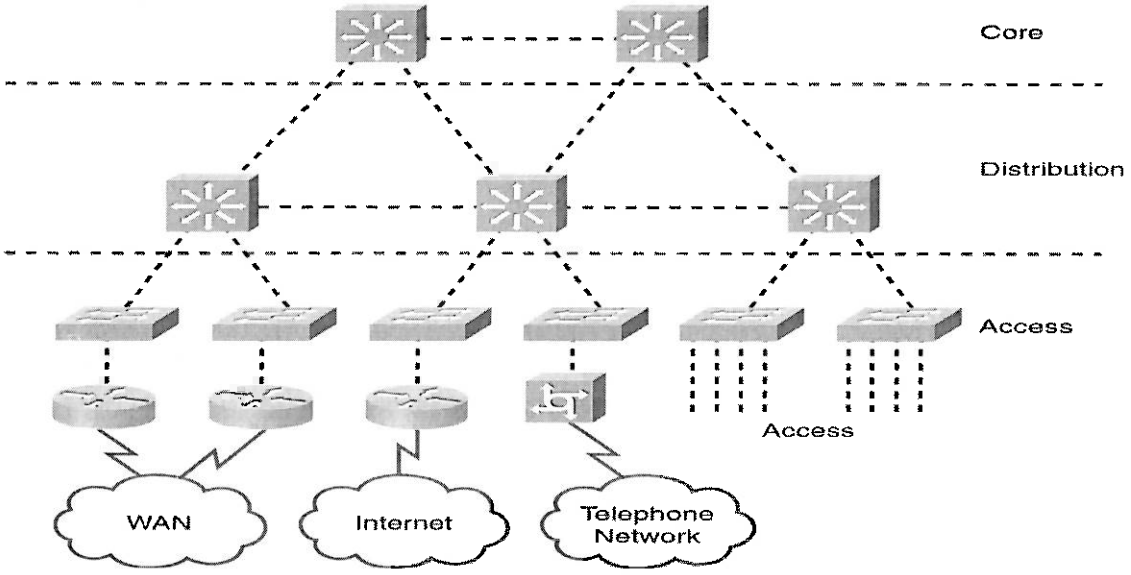
e) A wide area network in any organization should facilitate remote access for business continuity. With the use of one (1) example, explain any two (2) protocols or software utilities an organization can use for remote access. (6 marks)

[Total: 40 marks]

SECTION B: Answer any three (3) questions in this section.

QUESTION 2

a) The Hierarchical Network Model provides a modular framework that allows flexibility in network design and facilitates ease of implementation and troubleshooting in the WAN infrastructure. The hierarchical network model includes the core, distribution and access layers. With the use of one (1) example of devices at each layer, describe these layers. (9 marks)



b) WAN operations focus primarily on Layer 1 and Layer 2. WAN access standards typically describe both physical layer delivery methods and data link layer requirements, including physical addressing, flow control, and encapsulation. Discuss the following WAN technologies:

i) Circuit switching (4 marks)
ii) Packet switching (4 marks)
iii) Cell switching (3 marks)

[Total: 20 marks]

QUESTION 3

a) An IP address is a numerical label used to identify a device on a network. However, there are three (3) methods used to assign IP addresses on devices in an organisation. Stating the circumstances were they are used, explain these methods (9 marks)

b) The topology of a network is the arrangement and relationship of the network devices and the interconnections between them. With the use of a diagram in each case, differentiate between a Point-to-point topology and Hub-and-spoke topologies. (6 marks)

c) You have just been employed at Zamnet as Network technician. You have been informed that one of your clients is complaining of network outages consistently. Considering that there are many problems that interrupt network services on a network, explain the troubleshooting process you would undertake to solve the problem. (5 marks)

[Total: 20 marks]

QUESTION 4

a) Data link layer protocols define how data is encapsulated for transmission to remote sites and the mechanisms for transferring the resulting frames. A variety of technologies are used, such as PPP, ISDN, Frame Relay, X25 and ATM. Differentiate between asynchronous transfer mode (ATM) and frame relay. (6 marks)

b) In point-to-point protocol (PPP), we can secure communication between two points using authentication protocols. There are two ways in which we can configure PPP authentication security, these are:

i) PAP (Password Authentication Protocol) (3 marks)

ii) CHAP (Challenge-Handshake Authentication Protocol) (5 marks)

Explain these authentication methods.

c) Write down the syntax for configuring the following in packet tracer.

i) PPP (2 marks)

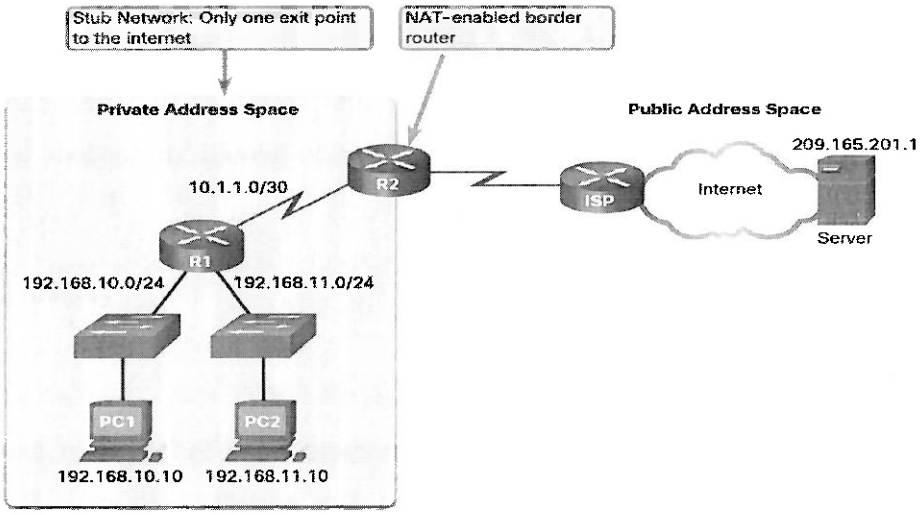
ii) PAP (2 marks)

iii) CHAP (2 marks)

[Total: 20 marks]

QUESTION 5

a) A Network address translation (NAT) router typically operates at the border of a public and private network as shown in the diagram below:



i) Describe two (2) uses of NAT on wide area networks (WANs) (4 marks)

ii) State the three (3) ranges of IP addresses used for NAT. (2 marks)

iii) Differentiate between Static and Dynamic NAT (6 marks)

b) Access Control Lists (ACLs) enable you to control traffic into and out of your network. An ACL can be as simple as permitting or denying network hosts, IP addresses or to control network traffic based on the TCP port being used. Write down an ACL that can do the following:

i) Allow traffic from all devices on network 192.168.15.0 and block all others

(2 marks)

ii) Deny file transfer protocol (FTP) traffic from network 10.0.0.0 (3 marks)

iii) Deny telnet traffic from network 130.65.0.0 (3 marks)

[Total: 20 marks]